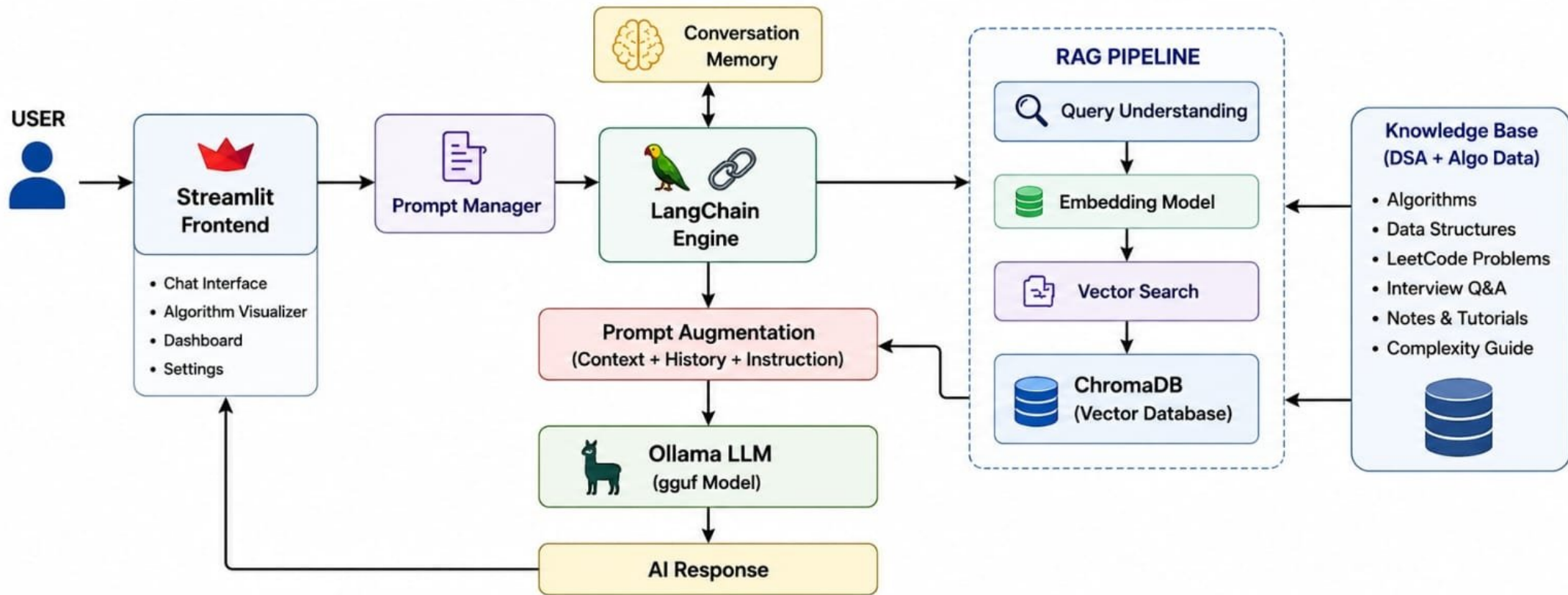


1. LLM SYSTEM ARCHITECTURE (OVERALL)



2. LLM INFRASTRUCTURE CONFIGURATION

DEVELOPMENT



- CPU : i5 / Ryzen 5
- RAM : 16 GB
- STORAGE : 100 GB SSD
- GPU : Optional
- OS : Ubuntu 22.04

FINE-TUNING



- GPU : RTX 3060 (12GB)
or T4 (16GB) or A100
- RAM : 32 GB+
- STORAGE : 200 GB+
- OS : Ubuntu 22.04

PRODUCTION



- CPU : 8 Core
- RAM : 32 GB
- GPU : RTX 4060 / 4070
- STORAGE : 500 GB SSD
- OS : Ubuntu 22.04

SOFTWARE STACK



Python
3.11



Ubuntu
22.04



Ollama
(LLM Runtime)



LangChain
(Framework)



ChromaDB
(Vector DB)

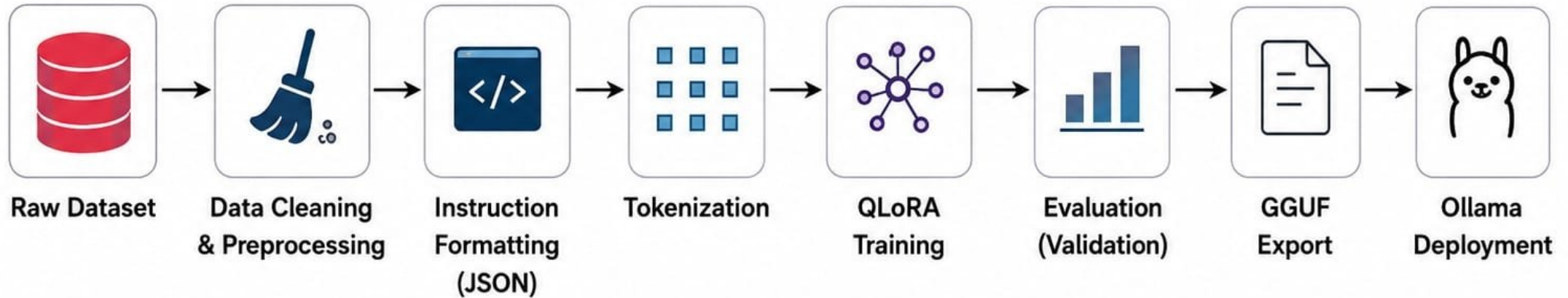


Streamlit
(Frontend)



Docker
(Deployment)

3. FINE-TUNING ARCHITECTURE



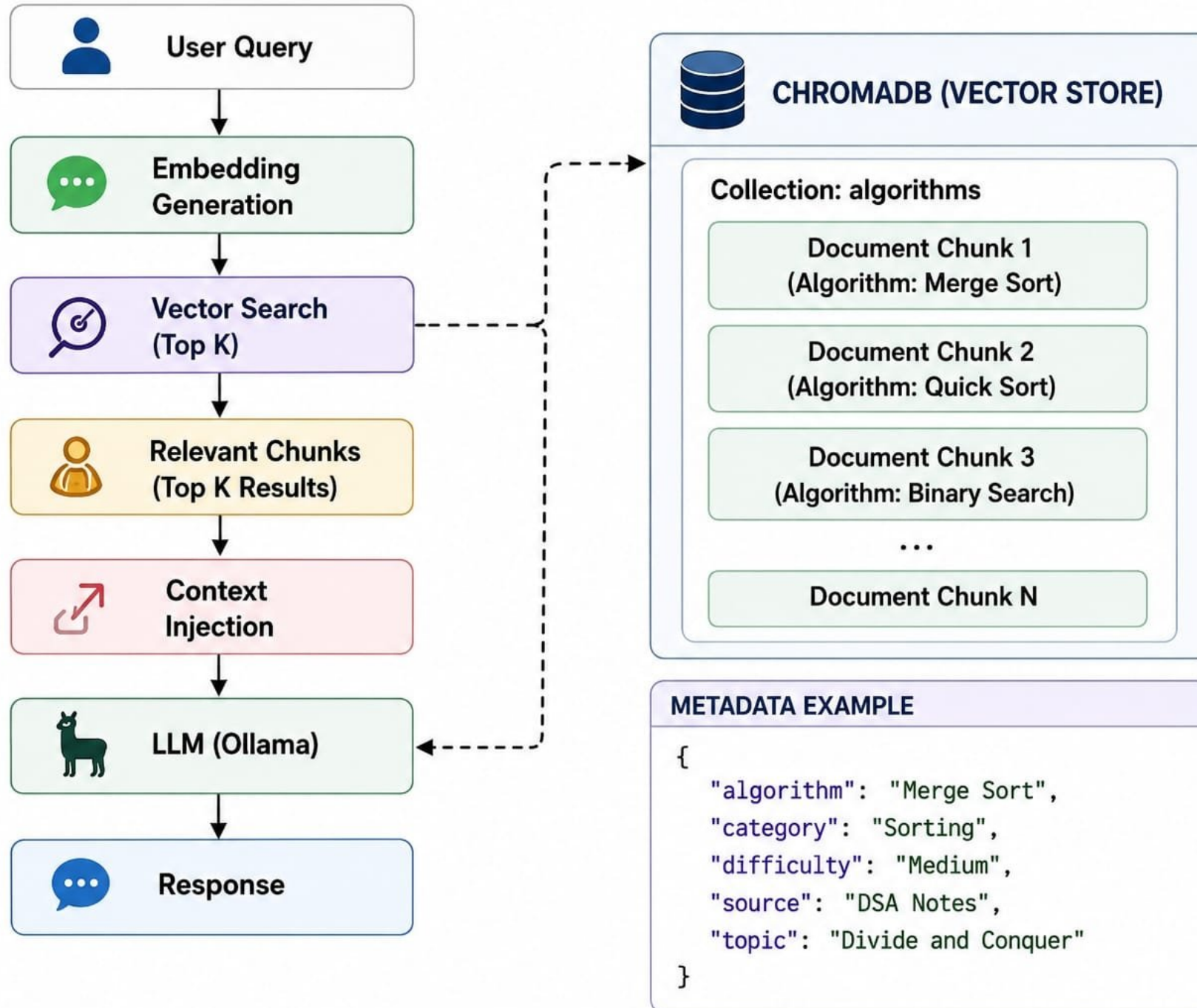
DATASET SOURCES



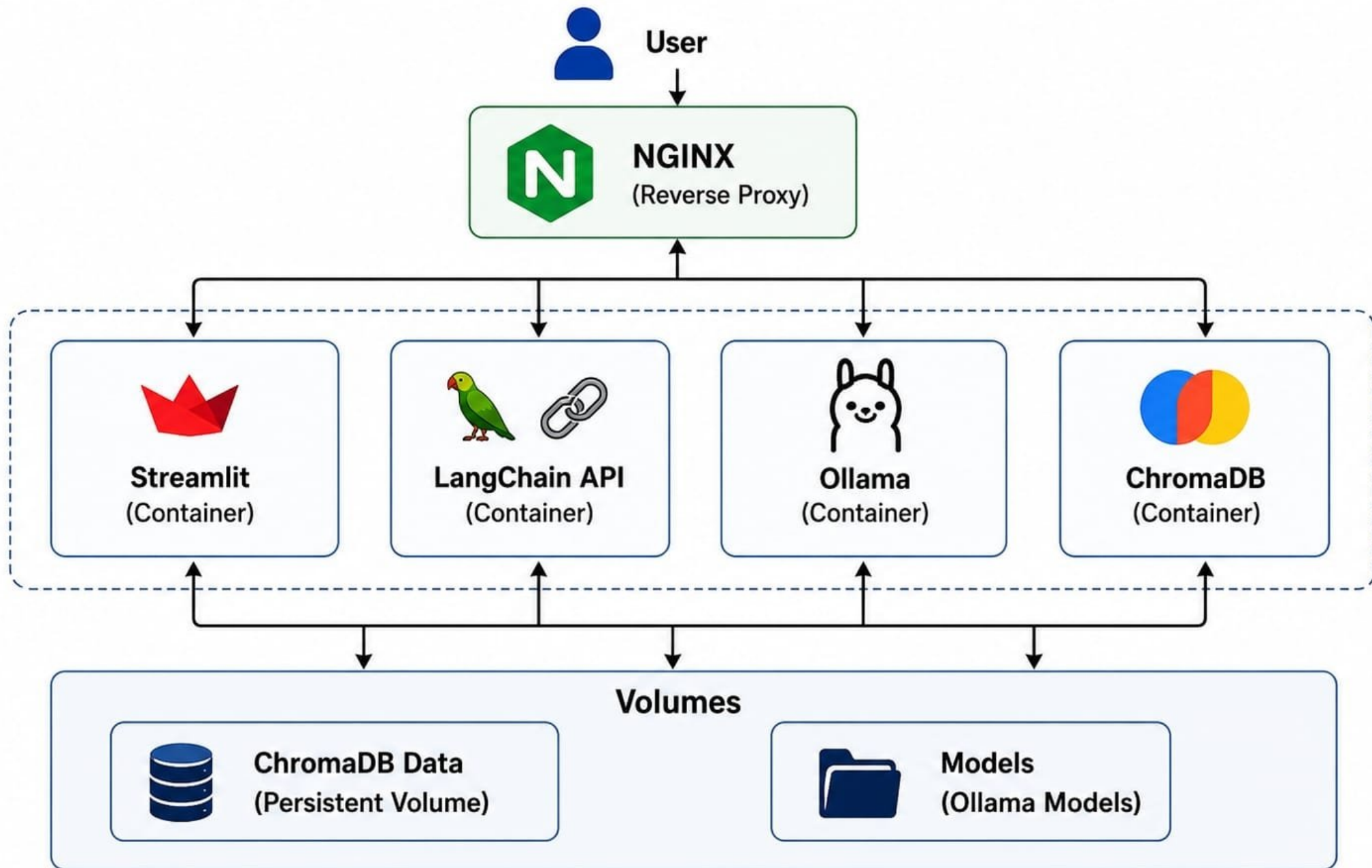
SAMPLE DATA FORMAT

```
{  
  "instruction": "Explain Binary Search",  
  "input": "",  
  "output": "Binary Search works on sorted  
             arrays by repeatedly dividing..."  
}
```


4. RAG PIPELINE ARCHITECTURE



5. DEPLOYMENT ARCHITECTURE (DOCKER)



6. OLLAMA CONFIGURATION



MODEL SELECTION

- Llama 3 8B (Recommended)
- Qwen 3 8B
- Mistral 7B

OLLAMA COMMANDS



```
ollama pull llama3:8b  
ollama serve  
ollama list  
ollama run llama3:8b
```

INFERENCE PARAMETERS (Recommended)



Temperature
0.2 – 0.3



Top-P
0.9



Top-K
40



Context Length
8192

7. LANGCHAIN CONFIGURATION

COMPONENTS



PromptTemplate



ConversationBufferMemory



Retriever (ChromaDB)

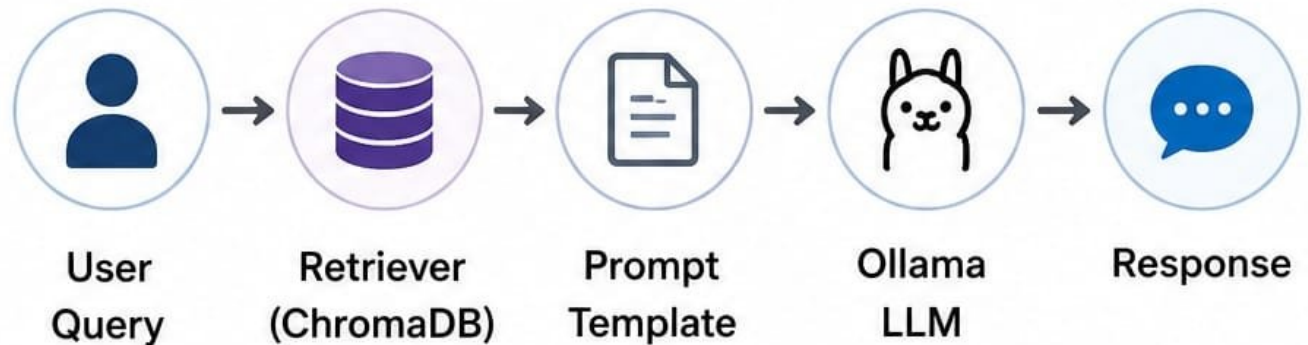


RetrievalQA Chain



OllamaLLM

CHAIN FLOW



- Stores previous conversation
- Maintains context
- Improves multi-turn interactions

8. PERFORMANCE & QUALITY TARGETS



Response Time
< 3 sec



Retrieval Time
< 500 ms



Embedding Time
< 300 ms



Accuracy
> 90%



Hallucination Rate
< 5%

MONITORING STACK



Prometheus



Grafana



Loki



Docker Stats

LOGGING



Application Logs



Error Logs



Access Logs